

Project Title

Library Book Request Management in ServiceNow

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Project Report

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Project Overview

The Library Book Request Management project in ServiceNow is developed to simplify and automate the process of requesting library books in a college or educational institution. Students frequently request books from the library for academic purposes. Managing these requests manually can result in delays, misplaced records, lack of proper tracking, and increased administrative effort.

This solution utilizes the ServiceNow platform to automate the complete lifecycle of library book requests. Students can submit their book requests through a custom application, where the system validates the submitted information, ensures that the issue date is not in the past, automatically generates a unique request number, stores the request in a custom table, and sends an email notification to the librarian whenever a new request is submitted.

The project integrates various ServiceNow components, including Custom Tables, Custom Fields, Auto Number Generation, Business Rules, Flow Designer, and Email Notifications, to build an efficient and reliable library request management system. By automating these activities, the solution reduces manual work, improves request tracking, enhances communication between students and librarians, ensures data accuracy, and provides a streamlined workflow for managing library book requests.

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1. Introduction :

In educational institutions, an efficient library management system plays a vital role in supporting students' academic learning and research activities. Students frequently request books from the college library for coursework, projects, examinations, and reference purposes. Managing these book requests manually often requires significant administrative effort and may result in delays, misplaced records, data entry errors, and difficulties in tracking request status.

As the number of students and library resources increases, handling book requests manually becomes increasingly inefficient. This can lead to delayed book issuance, poor communication between students and librarians, limited visibility into request progress, and inconsistencies in maintaining request records. An automated solution helps streamline the entire process, ensuring that book requests are processed accurately and efficiently.

The **Library Book Request Management** project, developed using the **ServiceNow** platform, provides a centralized and automated solution for managing library book requests. Students can easily submit book requests through a custom application, while automated workflows validate the submitted information, ensure that the issue date is not in the past, automatically generate a unique request number, store the request details in a custom table, and send email notifications to the librarian whenever a new request is submitted. This automation minimizes manual intervention and improves the overall efficiency of the library request process.

The proposed system offers the following benefits:

- Faster processing of library book requests.
- Standardized and consistent request management.
- Reduced manual intervention and administrative workload.
- Improved accuracy through automatic validation of issue dates.
- Automatic generation of unique request numbers for easy tracking.
- Enhanced communication through email notifications to the librarian.
- Better monitoring and tracking of all submitted book requests.
- Increased student satisfaction through a simple and efficient request process.

By utilizing the low-code capabilities of **ServiceNow**, this project demonstrates how workflow automation can simplify routine library operations while improving efficiency, accuracy, transparency, and record management. The solution also provides a scalable foundation that can be enhanced in the future by incorporating features such as book availability tracking, approval workflows, return reminders, and integration with the institution's library management system.

2. Project Objectives :

The main objective of this project is to develop an automated **Library Book Request Management System** using the ServiceNow platform that reduces manual effort and improves the efficiency of handling library book requests. The system is designed to provide students with a simple and centralized platform for submitting book requests, while automating the processes of validation, record generation, notifications, and request tracking.

The project also focuses on building a standardized workflow that enhances library service quality, increases operational efficiency, and ensures timely processing of book requests with minimal human intervention.

The specific objectives of this project are:

- Develop a centralized platform for students to submit **library book requests**.
- Design user-friendly request forms to capture student and book details.
- Implement **validation to ensure that the Issue Date** is not entered as a past date.
- **Create a Custom Table** to securely store and manage library book request records.
- Configure **Auto Number Generation** to create unique request numbers automatically.
- Configure **Flow Designer** to automate the request submission workflow.
- Generate **email notifications** automatically whenever a new book request is submitted.
- Maintain accurate records for monitoring and tracking all submitted book requests.
- Reduce request processing time and improve communication between students and librarians.
- Build a scalable and flexible solution that can be extended in the future with features such as approval workflows, book availability tracking, return reminders, and integration with a library management system.

3. Key Features

The proposed **Library Book Request Management** system includes the following major features:

- User-friendly request form for students to submit library book requests.
- Custom table for centralized storage and management of library book request records.
- Automatic generation of unique **Request Numbers** for every submitted request.
- Validation of the **Issue Date** using a **Business Rule** to prevent selecting past dates.
- User-friendly form for capturing student and book details.
- Automated workflow using **Flow Designer** for processing new book requests.
- Automatic email notifications sent to the librarian when a request is submitted.
- Secure storage of all request information in the custom table.
- Real-time tracking of submitted library book requests.
- Improved communication between students and librarians through automated notifications.
- Scalable workflow architecture that can be extended with additional library management features.
- Reduced manual effort and faster processing of library book requests.
- Improved data accuracy through automated validation and request numbering.
- Efficient record management for monitoring and reporting library requests.

4. Project Milestones

The project implementation follows five major milestones.

Milestone 1: Custom Table and Field Creation

Objective

Create a centralized database to store and manage library book request records.

Activities:

- Identify the information required for library book requests.
- Create the Library Book Request custom table.
- Create all required fields (Student Name, Book Name, Author Name, Issue Date, Return Date, Status, Librarian Email, Request Number).
- Configure appropriate field data types.
- Design a structured database for storing request records.

Milestone 2: Form Configuration

Objective

Design a user-friendly form to collect all required library book request information.

Activities:

- Configure the form layout.
- Add all required fields to the request form.
- Arrange the fields in a logical order.
- Configure mandatory fields.
- Improve the overall usability of the request form.

Milestone 3: Business Rule and Flow Automation

Objective

Automate the validation and notification process for library book requests.

Activities:

- Create a Business Rule to validate the Issue Date.
- Prevent users from selecting past dates.
- Configure Auto Number Generation for Request Numbers.
- Create a Flow Designer flow.
- Configure the trigger for record creation.
- Add the Send Email action.
- Send automatic email notifications to the librarian.

Milestone 4: Testing

Objective

Validate the complete functionality of the Library Book Request Management application.

Activities:

- Test custom table creation.
- Test form submission.
- Test Business Rule validation.

- Test auto-generated Request Numbers.
- Test Flow Designer execution.
- Test email notifications.
- Perform end-to-end application testing.
- Conduct User Acceptance Testing (UAT).

Milestone 5: Deployment

Objective

Deploy the Library Book Request Management application for end users.

Activities:

- Verify all configurations.
- Publish the application.
- Monitor request creation and processing.
- Verify email notifications.
- Train library staff and end users.
- Monitor application performance and resolve any issues after deployment.

5. ServiceNow Developer Setup

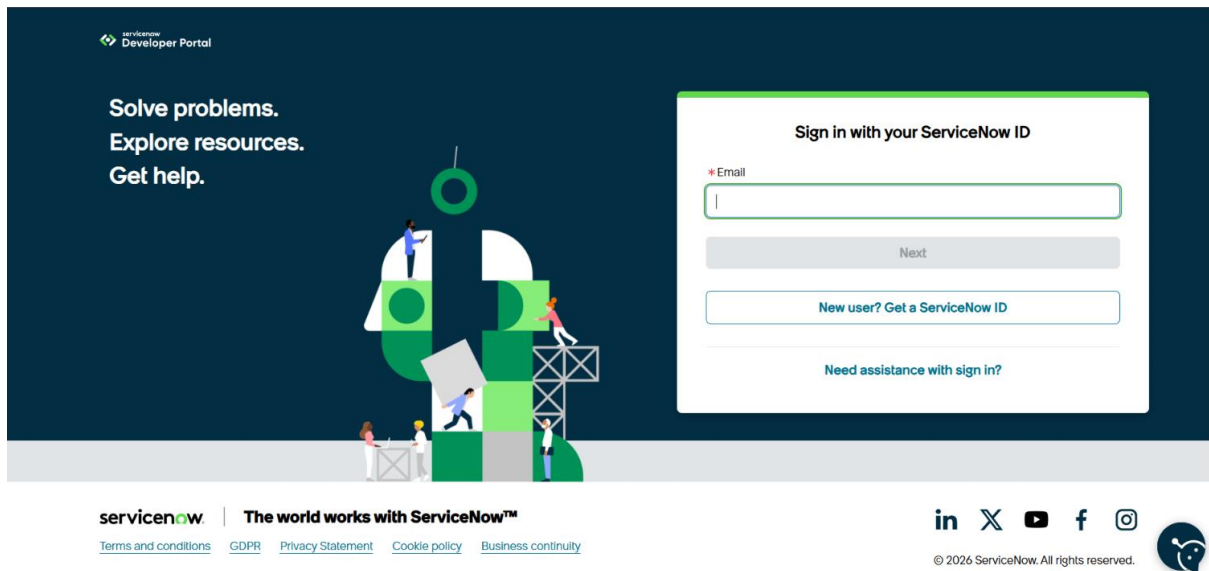
Before developing the **Library Book Request Management** application, a **ServiceNow Personal Developer Instance (PDI)** must be created. The PDI provides a fully functional ServiceNow environment where applications can be developed, tested, and deployed without affecting production systems.

The Personal Developer Instance provides access to all the ServiceNow features required for this project, including **Custom Tables, Form Configuration, Business Rules, Flow Designer, Auto Number Generation, and Email Notifications**. It allows developers to build, test, and validate the application in a safe environment before deployment.

5.1 Creating a ServiceNow Developer Account :

To begin development, perform the following steps:

1. Visit the ServiceNow Developer Portal.
2. Click **Sign Up** to create a free developer account.
3. Enter the required information:
 - First Name
 - Last Name
 - Email Address
 - Password
4. Verify the email address using the activation link sent to your registered email.
5. Log in to the ServiceNow Developer Portal.
6. Click **Start Building**.
7. Request a **Personal Developer Instance (PDI)**.
8. Wait for the instance to be provisioned.
9. Launch the instance and log in using the generated credentials.



5.2 Accessing Creator Studio

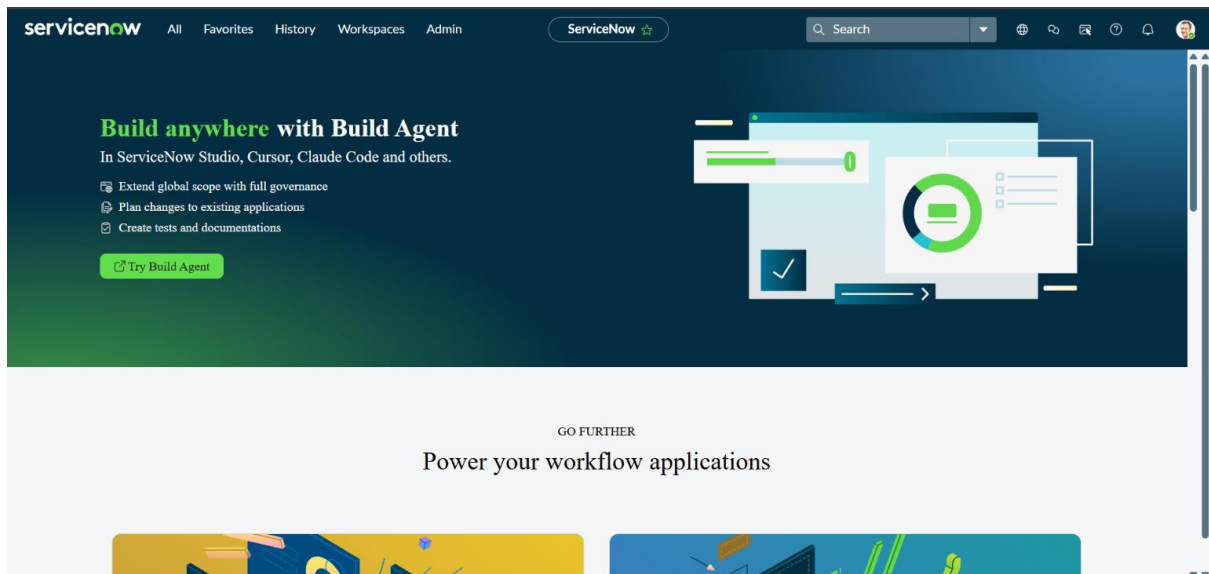
Once the **Personal Developer Instance** is ready, ServiceNow redirects the developer to **Creator Studio**.

Creator Studio provides a **no-code** and **low-code** environment for building applications. It simplifies application development by allowing developers to create custom tables, forms, workflows, automation, and integrations through an intuitive graphical interface.

For the **Library Book Request Management** project, Creator Studio is used to create the custom application, configure tables and fields, implement Business Rules, design workflows using Flow Designer, and automate email notifications.

Creator Studio offers features such as:

- Application Creation
- Data Modeling
- Custom Table Creation
- Form Builder
- Business Rule Configuration
- Flow Automation
- Auto Number Configuration
- Email Notification Configuration
- Workflow Configuration
- Application Testing and Deployment



6.1 Creation of Custom Table

A **Custom Table** was created in ServiceNow to maintain all **Library Book Request** records in a centralized database. This table stores the complete information related to each book request submitted by students, making it easier for librarians to manage, monitor, and track all requests efficiently.

The custom table serves as the primary data storage for the application and supports automation through **Business Rules**, **Flow Designer**, **Auto Number Generation**, and **Email Notifications**. It also provides flexibility for future enhancements such as approval workflows, book availability tracking, and return management.

Table Details

Table Name: Library Book Request

Purpose:

To store and manage all submitted library book requests in a centralized location for tracking, reporting, and administrative purposes.

Steps to Create the Custom Table

1. Open the **Application Navigator**.
2. Navigate to:

System Definition → Tables

3. Click **New** to create a new table.
4. Enter the required details:
 - **Label:** Library Book Request
 - **Table Name:** Library Book Request
 - **Application:** Global (or your application)
5. Enable the **Auto Number** option (if required).
6. Click **Submit** to create the custom table successfully.

Table

Library Book Request

✖ Label

Library Book Request

✖ Name

u_library_book_request

Employee Name

Department

IP Address

Device Name

Network Type

Location

Justification

Columns

Controls

Application Access

Table Columns

for text

Search

Dictionary Entries

Column label	Type	Reference	Max length	Default value	Display
Student Name	Reference	User	32		false
Updated	Date/Time	(empty)	40		false
Number	String	(empty)	40	javascript:global.getNextObjNumberPadded();	false
Created by	String	(empty)	40		false
Librarian Email	Reference	User	32		false
Issue Date	Date	(empty)	40		false
Status	Choice	(empty)	40		false
Return Date	Date	(empty)	40		false
Created	Date/Time	(empty)	40		false
Book Name	String	(empty)	100		false
Request Number	String	(empty)	100		false
Author Name	String	(empty)	100		false
Updated by	String	(empty)	40		false

Purpose of the Custom Table

The **Library Book Request** table serves as the central repository for storing all library book request records submitted by students. Each request is stored with details such as:

- Request Number
- Student Name
- Book Name
- Author Name
- Issue Date
- Return Date
- Status
- Librarian Email

The custom table provides a structured and organized way to manage all book requests. It enables librarians to easily view request details, track request status, and maintain accurate records. The table also supports automated workflows, business rule validation, and email notifications, reducing manual effort and improving the efficiency of the library request management process.

By maintaining all request information in a dedicated custom table, the application ensures better record management, improved tracking, enhanced data accuracy, and a scalable foundation for future enhancements.

6.2 Create Fields

Fields are created to collect the necessary information from students when they submit a **Library Book Request**. These fields appear on the Library Book Request form and capture all the information required for processing and managing the request.

The information entered into these fields is later utilized by **Business Rules, Auto Number Generation, Flow Designer, and Email Notifications** to automate the library book request process.

Steps to Create Fields

Follow the steps below to create the required fields:

1. Open the **Application Navigator**.
2. Navigate to:

System Definition → **Tables**

3. Open the Library Book Request table.
4. Scroll down to the Columns related list.
5. Click the New button to **create a new field**.
6. Configure the required properties, including:
 - Column Label
 - Column Name
 - Field Type
 - Maximum Length (if applicable)
 - Mandatory Status
 - Reference Table (for Reference fields)
7. Click Submit to create the field.
8. Repeat the above procedure until all the required fields have been created in the Library Book Request table.

Fields Created

Field Name	Field Type
------------	------------

Request Number	Auto Number
----------------	-------------

Student Name	Reference (User)
--------------	------------------

Book Name	String
-----------	--------

Author Name	String
-------------	--------

Issue Date	Date
------------	------

Return Date	Date
-------------	------

Status	Choice
--------	--------

Librarian Email	Email
-----------------	-------

Purpose of the Fields

The fields created in the Library Book Request table collect all the information required to process a library book request. The Student Name field stores the student's details, while the Book Name and Author Name fields identify the requested book. The Issue Date and Return Date fields specify the borrowing period. The Status field tracks the progress of the request, and the Librarian Email field is used by Flow Designer to send automatic email notifications. The Request Number is generated automatically through Number Maintenance, ensuring that every request has a unique identifier for tracking and management.

This field configuration provides a structured and efficient method for storing, processing, and managing library book requests within the ServiceNow platform.

The screenshot shows the 'Dictionary Entry' configuration page for the 'Student Name' field. At the top, a blue informational banner explains that dictionary entries manage how ServiceNow stores data in tables and fields, and provides instructions on selecting a table, field type, column label, and column name. Below this, the configuration form is divided into several sections:

- Table and Type:** The 'Table' is set to 'Library Book Request [u_library_book_requ...]' and the 'Type' is set to 'Reference'.
- Column Label and Name:** The 'Column label' is 'Student Name' and the 'Column name' is 'u_student_name'.
- Application and Properties:** The 'Application' is 'Global'. Properties include 'Active' (checked), 'Function field' (unchecked), 'Read only' (unchecked), 'Mandatory' (unchecked), and 'Display' (unchecked).
- Default Value:** A section titled 'The Default value specifies what value the field has when first displayed.' with an empty text input field.
- Related Links:** A list of links including 'Show Table', 'Run Point Scan', 'Advanced View', and 'SNY Links Versions (0)'.
- Access Controls:** A section with tabs for 'Access Controls', 'Choices', 'Attributes', and 'Labels (1)'. It includes a search bar and a table with columns: Name, Decision Type, Operation, Type, Active, Updated by, and Updated.

Field Configuration

The Library Book Request table consists of different types of fields to collect all the information required for processing a library book request. Each field is selected based on the type of data that needs to be captured from the student.

Field Name	Field Type
Student Name	Reference (User)
Book Name	String
Author Name	String
Issue Date	Date
Return Date	Date
Status	Choice
Librarian Email	Email
Request Number	Auto Number

Types of Fields Used

1. Reference Field

A Reference field is used to retrieve data from another ServiceNow table by referencing an existing record.

Example:

- Student Name → **User (sys_user)** Table

Purpose:

This field enables the system to select an existing user as the student, reducing manual data entry and improving data accuracy.

2. String Field

A String field is used to capture short text values entered by the user.

Examples:

- Book Name

- Author Name

Purpose:

These fields allow students to enter the name of the book and its author for the library request.

3. Date Field

A Date field is used to allow users to select a valid date from the calendar.

Examples:

- Issue Date
- Return Date

Purpose:

The Issue Date is validated using a Business Rule to ensure that students cannot select a past date. The Return Date specifies the expected date for returning the borrowed book.

4. Choice Field

A Choice field allows users to select one option from a predefined list of values.

Example:

Status

Available options include:

- Requested
- Approved
- Rejected
- Issued
- Returned

Purpose:

This field indicates the current status of the library book request and helps the librarian monitor and manage the request throughout its lifecycle.

5. Email Field

An Email field is used to capture a valid email address.

Example:

- Librarian Email

Purpose:

This field stores the librarian's email address so that Flow Designer can automatically send an email notification whenever a new library book request is submitted.

6. Auto Number Field

An Auto Number field is used to automatically generate a unique request number for every library book request.

Example:

- Request Number

Generated values include:

- LIB1000
- LIB1001
- LIB1002

Purpose: The Request Number uniquely identifies each library book request, making it easier to track, search, and manage records. The number is generated automatically through Number Maintenance, eliminating manual numbering and ensuring uniqueness.

6.3 Dictionary Entry Configuration

The Dictionary Entry in ServiceNow defines the properties and behavior of a field within a table. In this project, the Student Name field was configured as a Reference type in the Library Book Request table. This configuration enables the system to retrieve student information from the **User (sys_user)** table, reducing manual data entry and improving data consistency. The Dictionary Entry allows developers to configure various field properties such as the field type, column label, column name, default value, mandatory status, read-only status, and display settings.

Steps to Configure the Dictionary Entry

1. Open the **Library Book Request** table.

2. Select the **Student Name field**.

3. Open the Dictionary Entry for the field.

4. Configure the following properties:

- Field Type: Reference
- Reference Table: User (sys_user)
- Column Label: Student Name
- Column Name: u_student_name

Configure additional properties such as:

- Mandatory
- Read Only
- Display
- Default Value (if required)

Click Update to **save** the configuration.

Dictionary Entry Form Configuration

The Configure Form option allows developers to customize the Dictionary Entry form by selecting which field properties are displayed. This makes it easier to manage field configurations during application development.

Using the Configure Form option, developers can add or remove properties from the Dictionary Entry form according to the project requirements.

Steps to Configure the Dictionary Entry Form

1. Open the **Dictionary Entry record**.
2. Click the Additional Actions (:) menu.
3. Navigate to:

Configure → **Form Layout**

4. Select the required fields from the Available list.
5. Move them to the Selected list.
6. Arrange the fields in the required order.
7. Click Save to apply the changes.
8. Purpose of Dictionary Entry Configuration

The Dictionary Entry configuration provides several advantages:

- Defines the data type and behavior of each field.
- Ensures data consistency across the application.
- Supports reference fields for automatic data retrieval.
- Allows fields to be configured as mandatory or read-only.
- Simplifies field management and customization.

- Improves data accuracy and user experience.
- Provides flexibility for future enhancements.

The screenshot shows the 'Dictionary Entry' configuration page for 'Student Name'. A context menu is open over the 'Table' field, showing options like 'Save', 'Insert', 'Export', 'View', 'Configure', etc. The 'Table' is set to 'Library Book Request [u_library_book_requ...]' and the 'Type' is 'Reference'. The 'Column label' is 'Student Name' and the 'Column name' is 'u_student_name'. The 'Default value' field is empty. The 'Form view and section' section shows 'Default view' selected. The 'Create new field' section shows 'Name' as 'Student Name', 'Type' as 'String', and 'Field length' as 'Small (40)'. The 'Access Controls' table is visible at the bottom.

Name	Decision Type	Operation	Type	Active	Updated by	Updated

6.4 Number Maintenance Configuration

Number Maintenance is used in ServiceNow to automatically generate a unique request number for every new record created in the Library Book Request table. This eliminates manual entry of request numbers and ensures that each request has a unique identifier for tracking and management.

In this project, Number Maintenance is configured to generate request numbers automatically with the prefix LIB, followed by a sequential number (for example, LIB1000, LIB1001, LIB1002, etc.).

Objective

The implementation of Number Maintenance is intended to achieve the following objectives:

- Automatically generate a unique Request Number.
 - Eliminate manual entry of request numbers.
 - Improve request tracking and record management.
 - Ensure every library book request has a unique identifier.
 - Reduce duplicate request numbers and human errors.
-

Steps to Configure Number Maintenance

1. Open the **Application Navigator**.
 2. Search for **Number Maintenance**.
 3. Navigate to: **System Definition** → Number Maintenance
 4. Click **New**.
 5. Enter the required details:
 - Table: Library Book Request
 - Prefix: LIB
 - Number: 1000
 6. Click Submit to **save** the Number Maintenance record.
 7. Create a new record in the Library Book Request table to verify that the request number is generated automatically.
-

Example

Whenever a new request is submitted, ServiceNow automatically generates request numbers such as:

- LIB1000
- LIB1001
- LIB1002
- LIB1003

Each request number is unique and is assigned automatically by the system.

Number New record

* Table Library Book Request

Prefix LIB

* Number 1,000

Application Global

Number of digits 7

Submit

Related Links

[Show Counter](#)

6.5 Business Rule Configuration

A Business Rule is used in ServiceNow to automate server-side processing whenever a record is inserted, updated, deleted, or displayed. In this project, a Business Rule is created to validate the Issue Date entered by the student.

The Business Rule ensures that students cannot select an Issue Date that is earlier than the current date. If a past date is entered, the system displays an error message and prevents the record from being saved. This validation helps maintain data accuracy and ensures that only valid library book requests are processed.

Objective

The implementation of the Business Rule is intended to achieve the following objectives:

- Validate the Issue Date before saving the record.
- Prevent users from selecting a past date.
- Improve data accuracy and consistency.
- Reduce invalid library book requests.
- Ensure compliance with library borrowing policies.

Steps to Create the Business Rule

1. Open the **Application Navigator**.

Navigate to:

System Definition → Business Rules

1. Click **New**.
2. Enter the following details:

Property Value

Name Validate Issue Date

Table Library Book Request

When Before

Insert Checked

Update Checked

Advanced Checked

5. Paste the following script.

Business Rule Script

```
(function executeRule(current, previous) {  
  
    var today = new GlideDate();  
    today.setValue(gs.nowDate());  
  
    if (current.issue_date < today) {  
        gs.addErrorMessage("Issue Date cannot be in the past.");  
        current.setAbortAction(true);  
    }  
  
})(current, previous);
```

6. Click Submit or Update to save the Business Rule.

Working of the Business Rule

- When a student submits a new library book request, the Business Rule executes before the record is saved.
- The rule compares the Issue Date entered by the student with the current system date.
- If the selected Issue Date is earlier than today's date, an error message is displayed:
 - "Issue Date cannot be in the past."
- The record is not saved until a valid Issue Date (today or a future date) is selected.
- If the Issue Date is valid, the record is saved successfully, and the Flow Designer process continues to send an email notification to the librarian.

Benefits of the Business Rule

- Prevents invalid Issue Dates.
- Improves data quality and accuracy.
- Enforces library request validation rules.
- Reduces manual verification by librarians.
- Ensures only valid book requests are processed.
- Integrates seamlessly with Flow Designer and Number Maintenance.

The screenshot displays the ServiceNow Business Rules configuration page. At the top, there's a header with 'Business Rules' and 'Active' tabs, along with a search bar and a 'New' button. Below this is a table listing various business rules. The table has columns for Name, Active, Table, Application, Order, and Updated. The rules listed include 'm2mSLABreakdownRestrictBreakdownDefs', 'queryForPortalPages', 'Group Manager Change', 'Set Interaction Context', 'approval query', 'Start publication workflow', 'getNodeInSameCluster', 'getResourceInSameCluster', 'Notify instance clone schedule server', 'get_lat_long', 'Create new UX Experience', 'Enforce Parent being in layer above', and 'Generate attributes record'.

Below the table, there's a section for the 'Validate Issue Date' rule. It shows the rule's name, table ('Library Book Request [u_library_book_req...]', application ('Global'), and status ('Active' and 'Advanced' are checked). The 'When to run' tab is selected, and the 'Script' tab is active, showing the following JavaScript code:

```

1 (function executeRule(current, previous) {
2
3     var today = new GlideDate();
4     today.setValue(gs.nowDate());
5
6     if (current.issue_date < today) {
7         gs.addErrorMessage("Issue Date cannot be in the past.");
8         current.setAbortAction(true);
9     }
10
11 })(current, previous);

```

At the bottom, there are 'Update' and 'Delete' buttons, and a 'Related Links' section.

6.6 Create Flow for Email Notification

Flow Designer is a low-code automation tool in ServiceNow that automates business processes without writing complex scripts. In this project, a flow is created to automatically send an email notification to the librarian whenever a new Library Book Request is submitted.

The flow is triggered when a new record is created in the Library Book Request table. It retrieves the request details and sends an email containing the request information to the librarian.

Objective

The implementation of the email notification flow is intended to achieve the following objectives:

- Automatically notify the librarian when a new book request is submitted.
- Eliminate manual email communication.
- Improve communication between students and the librarian.
- Ensure timely processing of library book requests.
- Automate the notification process using Flow Designer.

Steps to Create the Flow

1. Open the **Application Navigator**.
2. Search for **Flow Designer**.
3. Navigate to:

Process Automation → Flow Designer

4. Click **New**.
5. Select **Flow**.

The screenshot shows the Workflow Studio interface. At the top, there's a navigation bar with tabs: Playbooks, Flows (selected), Subflows, Triggers, Actions, and Decision tables. A notification banner at the top says "New version available: Upgrade to get the latest Workflow Studio features. Learn more". Below the tabs, there's a "Flows" section with a "100" count and "Last refreshed just now." Below this is a table of flows.

Name	Application	Status	Active	Updated	Updated by
Application Intake Request Flow (Deprecated)	Application Intake	Published	false	2026-04-20 11:37:47	system
Application Intake Request V2	Application Intake	Published	true	2026-04-20 11:37:42	system
Attendance status	Altam servicenow	Published	true	2026-05-25 23:44:21	admin
Attendance status	Global	Published	true	2026-05-26 00:04:30	admin
Benchmark Recommendation Evaluator	Benchmarks Spoke	Published	true	2026-02-02 20:07:22	system
Business process approval flow	Global	Published	true	2020-09-27 22:06:13	admin
Change - Cloud Infrastructure - Authorize	Global	Published	true	2020-11-11 07:08:05	admin
Change - Conflict Detection	Global	Published	true	2026-02-02 19:59:23	system
Change - Emergency - Authorize	Global	Published	true	2020-10-06 05:39:49	admin
Change - Emergency - Implement	Global	Published	true	2020-09-23 05:06:26	admin
Change - Emergency - Review	Global	Published	true	2020-10-27 04:18:08	admin
Change - Normal - Assess	Global	Published	true	2020-10-06 05:37:05	admin
Change - Normal - Authorize	Global	Published	true	2020-10-06 05:38:35	admin
Change - Normal - Implement	Global	Published	true	2020-09-23 04:23:59	admin
Change - Standard	Global	Published	true	2020-09-23 05:06:01	admin

On the right side, there's a "New" dropdown menu open, showing options: Playbook, Flow (highlighted), Subflow, Trigger, Action, Decision table, and Help me decide. Below this, there's a "Latest updates" section with a list of recent changes, including "System Administrator modified Library booknotification" and "System Administrator modified Library Book Notification".

6. Enter the following details:

Property	Value
Flow Name	Library Book Notification
Application	Your Application / Global

Workflow Studio Library Book Notification Flow • Global New Flow Flow • None

Let's get the details for your flow

Flow name *

Application *

Description

[Show additional properties](#)

Cancel Build flow

7. Click Submit.

Configure the Trigger

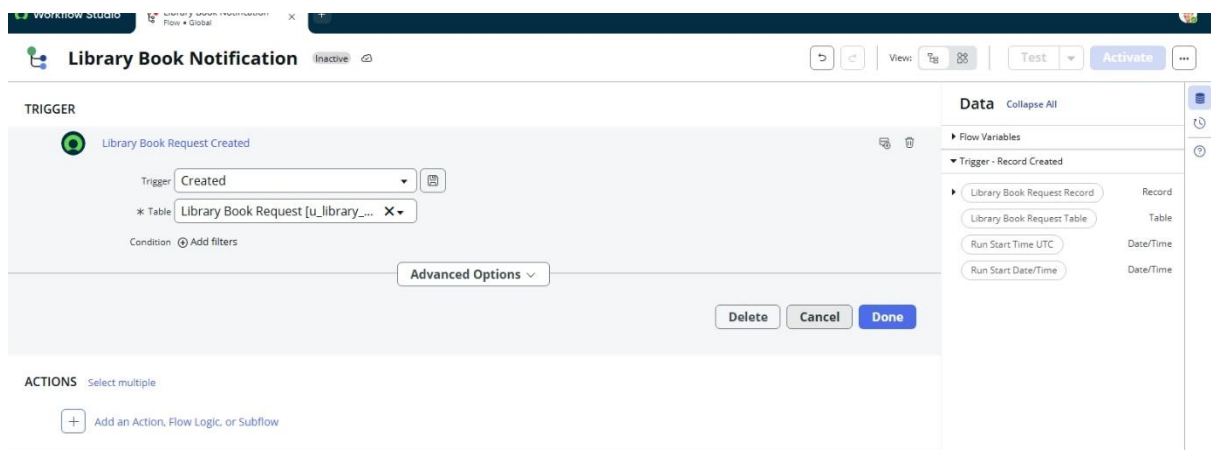
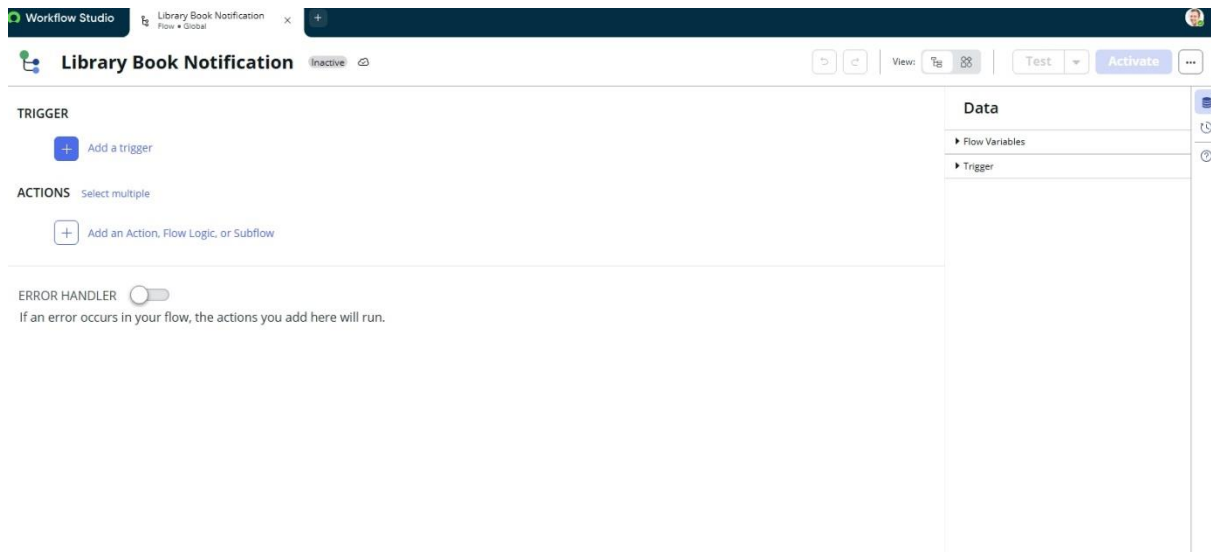
1. Click **Add Trigger**.
2. Select **Record Created**.
3. Configure the trigger:

Property	Value
----------	-------

Table	Library Book Request
-------	----------------------

4. Click **Done**.

The flow will now execute automatically whenever a new Library Book Request record is created.



Add Send Email Action

1. Click + **Add Action**.
2. Search for Send Email.
3. Select the **Send Email** action.
4. Configure the email details.

Property Value

To Librarian Email

Subject New Library Book Request

Body Dynamic request details

Email Body

Hello Librarian,

A new library book request has been submitted.

Request Number: LIB001003

Student Name: Harika

Book Name: Java Programming

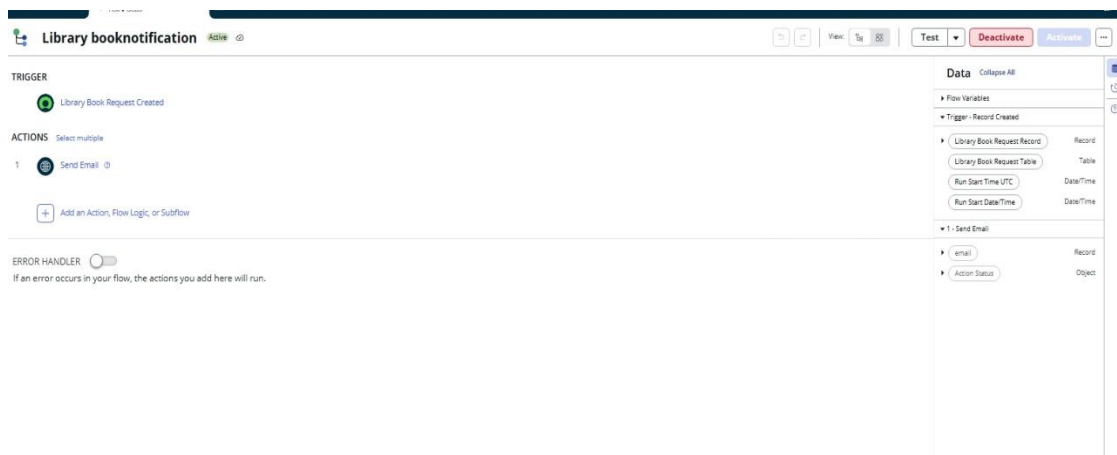
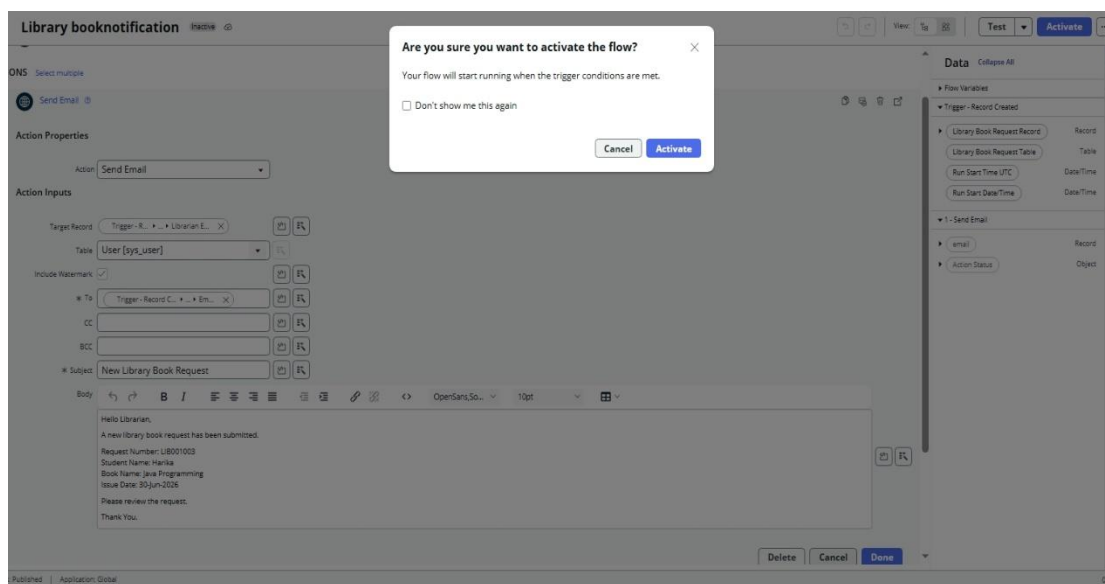
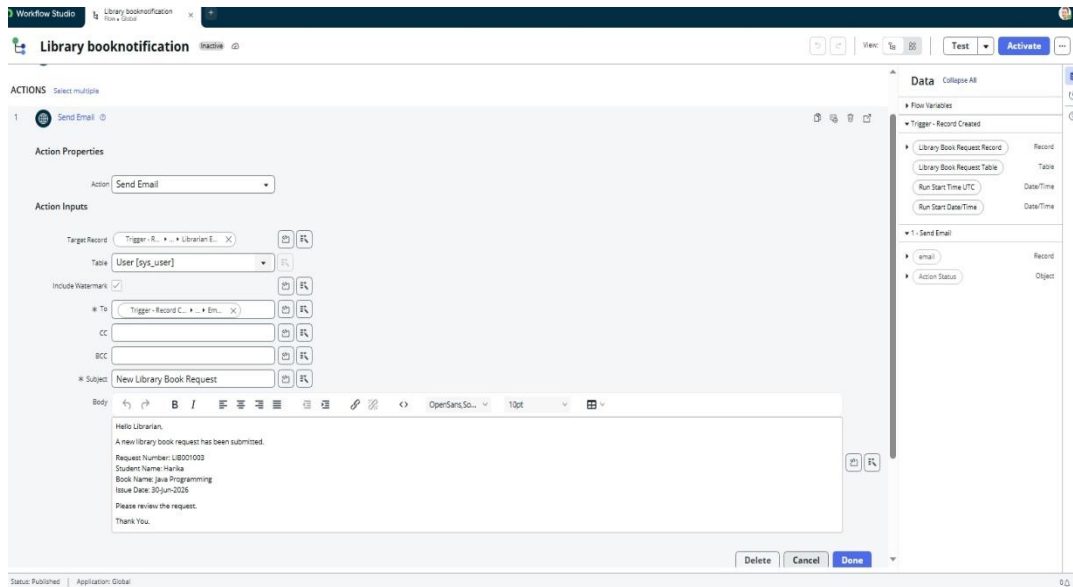
Issue Date: 30-Jun-2026

Please review the request.

Thank You.

Publish the Flow

1. Click **Save**.
2. Click Publish.
3. The flow is **now active** and ready for use.



Working of the Flow

When a student submits a Library Book Request, the Record Created trigger automatically starts the flow. The flow retrieves the request details from the newly created record and executes the Send Email action. An email containing the Request Number, Student Name, Book Name, and

Issue Date is sent to the librarian. This automation ensures that the librarian is immediately informed about new book requests, reducing manual communication and improving the efficiency of the library request process.

Benefits of Flow Designer

- Automates email notifications.
- Eliminates manual communication.
- Provides real-time notifications.
- Improves communication between students and the librarian.
- Reduces manual effort.
- Ensures timely processing of book requests.
- Enhances workflow efficiency.
- Supports future automation enhancements.

6.7 Testing the Application

After completing the configuration of the custom table, fields, Business Rule, Number Maintenance, and Flow Designer, the application must be tested to verify that all functionalities work as expected.

Steps to Test the Application

Step 1: Open the Form

1. Open the **Application Navigator**.

2. Navigate to:

Library Book Request

3. Click New to create a new library book request.

Step 2: Enter Sample Data

Fill in the form with the following sample values:

Field	Example Value
-------	---------------

Student Name	Harika
--------------	--------

Book Name	Java Programming
-----------	---------------------

Author Name	Herbert Schildt
-------------	-----------------

Issue Date	Tomorrow's Date
------------	--------------------

Return Date	Next Week's Date
-------------	---------------------

Status	Requested
--------	-----------

Librarian Email	librarian@colle ge.com
-----------------	---------------------------

After entering all the required details, click Submit

The screenshot shows a web form titled "Library Book Request" with the ID "LIB0001003". The form contains the following fields:

- Number: LIB0001003
- Request Number: 15
- Student Name: Abraham Lincoln
- Book Name: Java Programming
- Author Name: Herbert Schildt
- Issue Date: 2026-06-30
- Status: -- None --
- Librarian Email: Abraham Lincoln
- Return Date: 2026-07-07

At the bottom of the form, there are "Update" and "Delete" buttons.

6.8 Verify the Email Was Sent

1. Submit the **Library Book Request** form.
2. Open **Application Navigator**.
3. Navigate to:
System Mail → **Emails**
4. Search for the subject:
New Library Book Request

5. Check the Type/State:

- Sent → Email was sent successfully.
- Send Ready → Waiting to be sent.
- Failed → Email was not sent.

Emails

Created

Search

Actions on selected rows...

New

All > Created on Today

Q

Created

Recipients

Subject

Type

Notification type

User ID

Search

Search

Search

Search

Search

Search

2026-06-29 00:00:10

admin@example.com

Daily job to fetch Email Indicator Data and Email Notifications created completed with error

send-ready

SMTP

(empty)

2026-06-29 01:00:10

ailen.mottern@example.com

Restocking Request For APC 42U 3100 SP2 NetShelter Rack

send-ready

SMTP

(empty)

2026-06-29 01:00:12

ailen.mottern@example.com

Restocking Request For Dell Inc. PowerEdge M710HD Blade Server

send-ready

SMTP

(empty)

2026-06-29 01:00:12

ailen.mottern@example.com

Restocking Request For Fujitsu 1TB Hybrid Solid State Drive

send-ready

SMTP

(empty)

2026-06-29 04:27:12

abraham.lincoln@example.com

New Library Book Request

send-ready

SMTP

(empty)

6.9 Screenshots of Output

New Library Book Request

Recipients

abraham.lincoln@example.com

Body

<html><head></head><body><p></p><p data-tnymcerooblock="">Hello Librarian.</p><p data-tnymcerooblock="">A new library book request has been submitted.</p><p data-tnymcerooblock="">Request Number: \${Request Number}</p><p data-tnymcerooblock="">Student Name: \${Student Name}</p><p data-tnymcerooblock="">Book Name: \${Book Name}</p><p data-tnymcerooblock="">Issue Date: \${Issue Date}</p><p data-tnymcerooblock="">Please review the request.</p><p data-tnymcerooblock="">Thank You.</p></div><div style="display:inline">Ref:MSG0002011_sDR4D4mqd3G1H9jWqoa</div></body></html>

Content type

Headers

Message-ID: <387064685.5.1782732432360@app128168.ord191.service-now.com>
X-ServiceNow-Source: FlowDesigner-0290181da1f48bd4aa52f98b216833f4
X-ServiceNow-Generated: true
X-ServiceNow-SysEmail-Version: 2

Update

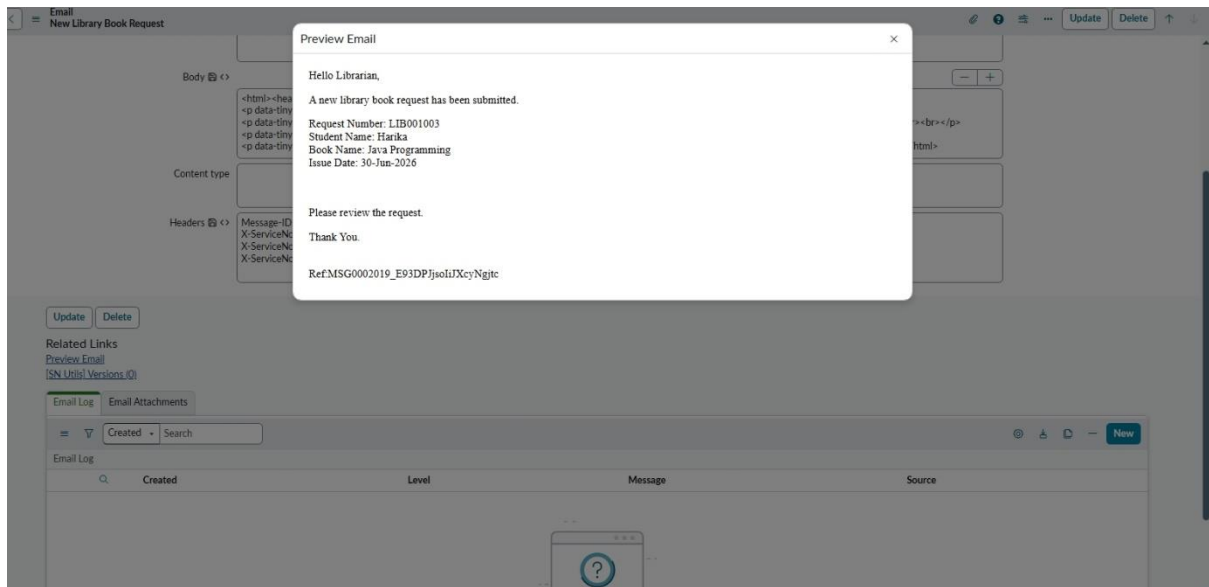
Delete

Related Links

[Preview Email](#)
[\[SN Utilis\] Versions \(0\)](#)

Email Log

Email Attachments



7. Conclusion

The Library Book Request Management System was successfully developed using the ServiceNow platform to automate the process of managing library book requests. The application provides a simple and centralized system that enables students to submit book requests efficiently while allowing librarians to manage and track requests with ease.

The project incorporates various ServiceNow features, including Custom Tables, Form Configuration, Number Maintenance, Business Rules, Flow Designer, and Email Notifications, to automate the complete request lifecycle. The Business Rule ensures that the Issue Date is valid, Number Maintenance automatically generates a unique request number for each request, and Flow Designer sends email notifications to the librarian whenever a new request is submitted.

By automating these processes, the system reduces manual effort, improves data accuracy, enhances request tracking, and ensures timely communication between students and librarians. The application also provides a structured and scalable solution that can be extended in the future with additional features such as approval workflows, book availability checking, due-date reminders, fine management, and integration with other library services.

Overall, the Library Book Request Management System demonstrates how the ServiceNow platform can be used to build an efficient, reliable, and user-friendly application that improves library operations and delivers better service to both students and library staff.